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Structural arrangement for attachment of a standoff insulator to an air core reactor

Abstract

An air core reactor including an improved structural arrangement for attaching cylindrical winding packages in the air core reactor is provided. Disclosed embodiments make use of spacer assemblies appropriately arranged to counterbalance bending moments that can develop during operation of the air core reactor and therefore improve the structural capability of the mechanical connection for the winding packages between respective spacers and spider units of the air core reactor.

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Background/Summary

BACKGROUND

(1) Disclosed embodiments relate generally to the field of electrical apparatuses, and, more particularly, to air core reactors.

BRIEF SUMMARY

(2) A disclosed embodiment is directed to an air core reactor including a first cylindrical winding package positioned to extend along a central axis from a first reactor end to a second reactor end, a second cylindrical winding package positioned radially inward of the first cylindrical winding package and positioned to extend along the central axis from the first reactor end to the second reactor end, a spider arm that extends in a direction away from the central axis along a radius to a spider end and is coupled to the first cylindrical winding package and the second cylindrical winding package, a first attachment bracket includes a first inner flange, a first outer flange, and a first web connecting the first inner flange to the first outer flange, the first inner flange coupled to the spider end, a first spacer positioned between the first cylindrical winding package and the second cylindrical winding package, the first spacer having a first spacer end coupled to the first outer flange of the first attachment bracket, and a first collar disposed around the first spacer end of the first spacer between the first attachment bracket and the first cylindrical winding package and the second cylindrical winding package.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

(2) 1. FIG. 1 is a view of an electrical apparatus, such as a dry-type air core reactor, that can benefit from disclosed structural arrangements for attaching conductor winding packages in the air core reactor.

(3) FIG. 2 illustrates an aspect of the subject matter in accordance with one embodiment.

(4) FIG. 3 illustrates an aspect of the subject matter in accordance with one embodiment.

(5) FIG. 4 illustrates an aspect of the subject matter in accordance with one embodiment.

DETAILED DESCRIPTION

(6) FIG. 1 is a view of an electrical apparatus, such as an air core reactor **100**, that can benefit from disclosed embodiments described in greater detail below. Disclosed embodiments involve an improved structural arrangement for attaching cylindrical winding packages in the air core reactor **100**. The terms air core reactor, air core inductor and air core coil are often used interchangeably by those skilled in the art and refer to inductors that involve an air core in lieu of a magnetic core made of a ferromagnetic material. An inductor (reactor, or coil) is a passive electrical component that may be used to store energy available in an electromagnetic field when electric current flows through the inductor.

(7) In the following detailed description, various specific details are set forth in order to provide a thorough understanding of such embodiments. However, those skilled in the art will understand that disclosed embodiments may be practiced without these specific details that the aspects of the present invention are not limited to the disclosed embodiments, and that aspects of the present invention may be practiced in a variety of alternative embodiments. In other instances, methods, procedures, and components, which would be well-understood by one skilled in the art have not been described in detail to avoid unnecessary and burdensome explanation.

(8) Furthermore, various operations may be described as multiple discrete steps performed in a manner that is helpful for understanding embodiments of the present invention. However, the order of description should not be construed as to imply that these operations need be performed in the order they are presented, nor that they are even order dependent, unless otherwise indicated.

Moreover, repeated usage of the phrase “in one embodiment” does not necessarily refer to the same embodiment, although it may.

(9) It is noted that disclosed embodiments need not be construed as mutually exclusive embodiments, since aspects of such disclosed embodiments may be appropriately combined by one skilled in the art depending on the needs of a given application.

(10) Air core reactor **100** includes one or more electrical devices, such as a plurality of radially-concentric, spaced-apart cylindrical winding packages **104**, **106** positioned about a central axis **108** that extends from a first reactor end **112** to a second reactor end **114**. The cylindrical winding packages **104**, **106** may define a centrally-disposed hollow cavity **120**. It will be appreciated that air core reactor designs may include fewer or substantially more winding packages than shown in FIG. 1 (e.g., ranging from one winding package to twenty or more winding packages). For simplicity of illustration, FIG. 1 illustrates just two winding packages labelled **104**, **106**.

(11) Without limitation, cylindrical winding packages **104**, **106** may be positioned between an upper spider unit **122** and a lower spider unit **124**, which, in certain embodiments, may function as terminals for connecting power lines and/or for interconnecting the cylindrical windings in a desired electrical configuration, such as a parallel circuit arrangement. It will be appreciated that the winding packages of the reactor are rigidly held between upper spider unit **122** and lower spider unit **124** by appropriate means, such as, without limitation, non-conductive materials, which may include precured composite members or uncured composite members (that are hardened when the reactor is cured). The spider units **122**, **124** may constitute structural members that facilitate lifting and/or fastening to the mounting system of a given reactor to other reactors, or both, or to a reactor foundation. Standoff insulator **128** may be attached to spider ends **118** and they may support current-carrying conductors **130**. Cylindrical winding packages **104**, **106** may be radially separated from one another by a plurality of circumferentially spaced-apart “duct sticks” or spacers **126**, which may be positioned to have a vertical orientation extending in a direction parallel to central axis **108**.

(12) The present inventors have recognized that the structural strength of certain known mounting

arrangements for the cylindrical winding packages in air core reactors **100** may be limited by the structural capability of a welded or bolted connection between the standoff insulator **128** and the spider units **122**, **126**. For example, mechanical loads can develop during operation of the air core reactor **100**, such as those forces that may result for heating and/or electromagnetic effects on spider arms **116** in the magnetic field of the air core reactor **100**. These loads can result in undesirable bending moments being formed and applied to the welded or bolted connection. These bending moments, if left unrestrained, as is generally the case in such known mounting arrangements, can potentially affect the reliability and durability of the welded connection between the spacers **126** and the spider units. The present invention innovatively increases the bending resistance over conventional connections, such as welded or bolted “T” brackets, by counteracting the bending moment with axial forces.

(13) At least in view of the foregoing recognition, disclosed embodiments make use of spacers **126** appropriately arranged to counterbalance such moments and therefore improve the structural capability of the mechanical connection between the spacers **126** and the spider units. Disclosed embodiments are also believed effective in resisting shear loads that may develop during operation of the air core reactor **100**. Without limitation, spacers **126** used in disclosed embodiments may be realized by way of appropriate adaptation of spacers **126**, generally used to facilitate convection cooling in the air core reactor **100**.

(14) In the embodiment shown in FIG. **1**, the air core reactor **100** includes a first cylindrical winding package **104** positioned to extend along a central axis **108** from a first reactor end **112** to a second reactor end **114**. A second cylindrical winding package **106** is positioned radially inward of the first cylindrical winding package **104** and positioned to extend along the central axis **108** from the first reactor end **112** to the second reactor end **114**. A spider arm **116** extends in a direction away from the central axis **108** along a radius **110** to a spider end **118** and is coupled to the first cylindrical winding package **104** and the second cylindrical winding package **106**. Although in FIG. **1** the arms of spider units **124**, **126** are illustrated as extending from central axis **108**, it will be appreciated that in certain embodiments, the spider arms **116** may be truncated. That is, the spider arms **116** need not extend from central axis **108** but from a point located between central axis **108** and spider end **118**.

(15) FIG. **2** is an isometric, fragmentary view of a first attachment bracket **102** of the air core reactor **100**. FIG. **3** shows a fragmentary top view first attachment bracket **102** of the air core reactor **100**. In an embodiment, the first attachment bracket **102** may include a first inner flange **204**, a first outer flange **202**, and a first web **216** connecting the first inner flange **204** to the first outer flange **202**. In an embodiment, the first inner flange **204** may be coupled to the spider end **118**. The first spacer **228** may be positioned between the first cylindrical winding package **104** and the second cylindrical winding package **106**, the first spacer **228** having a first spacer end **206** coupled to the first outer flange **202**.

(16) Without limitation, in the illustrated embodiment, the first attachment bracket **102** may be coupled to spider arm **116**, which may be part of an upper spider unit **122** (FIG. **1**). It will be appreciated that in certain applications, another attachment bracket **136** could be coupled to a second spider arm **134** of a second air core reactor **132** in a stacked embodiment, or any spider arm **116** of an air core reactor **100**.

(17) A first collar **208** may be disposed around the first spacer end **206** of the first outer flange **202** between the first attachment bracket **102** and the first cylindrical winding package **104** and the second cylindrical winding package **106**. In one aspect, first collar **208** is formed from a non-conductive or dielectric material, that is, made of a material having electrical insulating properties.

(18) In one non-limiting embodiment, the first outer flange **202** of the first attachment bracket **102** may include a first outermost surface **210** and the first inner flange **204** of the first attachment bracket **102** may include a first innermost surface **306**. In one non-limiting embodiment, the first spacer end **206** is disposed against the first outermost surface **210**. Alternatively, the first spacer

end **206** may be disposed against a surface opposite the first outermost surface **210** of the first outer flange **202**. In another embodiment depicted in FIG. 4, a third spacer **402** may be positioned between the first cylindrical winding package **104** and the second cylindrical winding package **106** and spaced circumferentially apart from the first spacer **202** by at least second outer flange **310** width **406**. The third spacer end **404** may be disposed against the inner surface **408**. The second collar **312** may extend around both spacer ends **304**, **206**. The air core reactor **100** may also include a first outer bolt assembly **212**, for example, comprising at least a nut and bolt, and may be used to affix the first outer flange **202** to the first outermost surface **210** of the first outer flange **202** of the first attachment bracket **102**.

(19) For the sake of simplicity of illustration, bolt assemblies are shown in connection with disclosed embodiments; it will be appreciated, however, that it is possible to use an alternative modality or combination of modalities to transfer compression and tensile loads, such as could include a geometric bearing feature (e.g., effective to transfer compression) and an adhesive (e.g., effective to transfer tension). For example, multiple separate bolt attachments may be used to affix the flanges **204** to the spacers **126** and/or spider ends **118**. In an embodiment, the first outer flange **202** may be extended in the direction of the central axis **108** such that it permits the fastening of more than one bolt assembly (such as **212**) thereto. In another embodiment, the first attachment bracket **102** may also include a first innermost surface **306** disposed against the spider end **118** and may include a first inner bolt assembly **214** to affix the first innermost surface **306** of the first inner flange **204** of the first attachment bracket **102** to the spider end **118**.

(20) In operation, when a bending moment **324** acts on the spider arm **116**, the first attachment bracket **102** opposes the bending moment **324** via its first outer flange **202** coupling to the first outer flange **202**. This arrangement provides an additional resistance force **326** force as opposed to being left unrestrained, as is the case in certain known arrangements.

(21) In one non-limiting embodiment, the first outer flange **202** of the first attachment bracket **102** is longitudinally aligned along a radius **110** of the air core reactor **100**. In another aspect, the first web **216** comprises a first web height **222** extending in a direction of the central axis **108**.

(22) In another aspect, the first inner flange **204** may include a first inner flange height **224** extending in a direction of the central axis **108**, wherein the first inner flange height **224** may be less than the first web height **222**. The first inner flange height **224** may be configured to accommodate a configuration of the spider end **118**, such as for fastener clearance. In another non-limiting embodiment, the first outer flange may include an outer flange length **322** extending in a direction of the radius, and the first inner flange **204** comprises an inner flange length **320** extending in a direction of the radius **110**, wherein the outer flange length **322** is greater than the inner flange length **320**. The inner flange length **320** may be configured to accommodate a configuration of the spider end **118**, such as for fastener clearance. In yet another aspect, the first outer flange **202** may include an angled edge portion **226** disposed away from the first web **216**.

(23) In another non-limiting embodiment, a third spacer **402**, having a third spacer end **404**, may be positioned between the first cylindrical winding package **104** and the second cylindrical winding package **106**. The third spacer **402** may be spaced circumferentially apart from the first spacer **228** by at least a first outer flange **202** width **406**. In an embodiment, the third spacer end **404** may be disposed against an inner surface **408** of the first outer flange **202**.

(24) The air core reactor **100** may also include a second attachment bracket **220** including a second inner flange **314**, a second outer flange **310**, and a second web **218** connecting the second inner flange **314** to the second outer flange **310**. In an aspect, the second inner flange **314** may include a second innermost surface **318**, the second innermost surface **318** coupled to the spider end **118** on a spider end **118** side **308** opposite the first attachment bracket **102**, a second spacer **302** positioned between the first cylindrical winding package **104** and the second cylindrical winding package **106** and circumferentially spaced apart from first outer flange **202**, the second spacer **302** having a second spacer end **304** coupled to a second outermost surface **316** of a second outer flange **310** of

the second attachment bracket **220**, and a second collar **312** disposed around the second spacer end **304** of the second spacer **302** between the second attachment bracket **220** and the first cylindrical winding package **104** and the second cylindrical winding package **106**. In another aspect, a second web **218** of the second attachment bracket **220** may be attached to the first web **216** along at least a portion of the first web height **222**.

(25) Therefore, disclosed embodiments are effective to counterbalance bending moments **324** and improve the structural capability of the mechanical connection between the spacer and the spider units. Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

LISTING OF DRAWING ELEMENTS

(26) **100** air core reactor **102** first attachment bracket **104** first cylindrical winding package **106** second cylindrical winding package **108** central axis **110** radius **112** first reactor end **114** second reactor end **116** spider arm **118** spider end **120** hollow cavity **122** upper spider unit **124** lower spider unit **126** spacers **128** standoff insulator **130** conductor **132** second air core reactor **134** second spider arm **136** another attachment bracket **202** first outer flange **204** first inner flange **206** first spacer end **208** first collar **210** first outermost surface **212** first outer bolt assembly **214** first inner bolt assembly **216** first web **218** second web **220** second attachment bracket **222** first web height **224** first inner flange height **226** angled edge portion **228** first spacer **302** second spacer **304** second spacer end **306** first innermost surface **308** side **310** second outer flange **312** second collar **314** second inner flange **316** second outermost surface **318** second innermost surface **320** inner flange length **322** outer flange length **324** bending moment **326** resistance force **402** third spacer **404** third spacer end **406** width **408** inner surface

(27) While embodiments of the present disclosure have been disclosed in exemplary forms, it will be apparent to those skilled in the art that many modifications, additions, and deletions can be made therein without departing from the scope of the invention and its equivalents, as set forth in the following claims.

Claims

1. An air core reactor comprising: a first cylindrical winding package positioned to extend along a central axis from a first reactor end to a second reactor end; a second cylindrical winding package positioned radially inward of the first cylindrical winding package and positioned to extend along the central axis from the first reactor end to the second reactor end; a spider arm that extends in a direction away from the central axis along a radius to a spider end and is coupled to the first cylindrical winding package and the second cylindrical winding package; a first attachment bracket comprising a first inner flange, a first outer flange, and a first web connecting the first inner flange to the first outer flange, the first inner flange coupled to the spider end; a first spacer positioned between the first cylindrical winding package and the second cylindrical winding package, the first spacer having a first spacer end coupled to the first outer flange of the first attachment bracket; and a first collar disposed around the first spacer end of the first spacer between the first attachment bracket and the first cylindrical winding package and the second cylindrical winding package.
2. The air core reactor of claim 1, wherein the first outer flange of the first attachment bracket includes a first outermost surface and the first inner flange of the first attachment bracket includes a first innermost surface.
3. The air core reactor of claim 2, wherein the first spacer end is disposed against the first outermost surface.
4. The air core reactor of claim 2, further comprising a first outer bolt assembly to affix the first spacer to the first outermost surface of the first outer flange of the first attachment bracket.
5. The air core reactor of claim 2, wherein the first innermost surface is disposed against the spider end.

6. The air core reactor of claim 5, further comprising a first inner bolt assembly to affix the first innermost surface of the first inner flange of the first attachment bracket to the spider end.
 7. The air core reactor of claim 1, wherein a bending moment applied to the spider arm is opposed by the first attachment bracket via its first outer flange coupling to the first spacer.
 8. The air core reactor of claim 1, wherein the first collar is an electrical insulator.
 9. The air core reactor of claim 1, wherein the first outer flange of the first attachment bracket is longitudinally aligned along a radius of the air core reactor.
 10. The air core reactor of claim 1, wherein the first web comprises a first web height extending in a direction of the central axis.
 11. The air core reactor of claim 10, wherein the first inner flange comprises a first inner flange height extending in a direction of the central axis.
 12. The air core reactor of claim 11, wherein the first inner flange height is less than the first web height.
 13. The air core reactor of claim 1, wherein the first outer flange comprises an outer flange length extending in a direction of a radius of the air core reactor and the first inner flange comprises an inner flange length extending in a direction of the radius, wherein the outer flange length is greater than the inner flange length.
 14. The air core reactor of claim 1, wherein the first outer flange comprises an angled edge portion disposed away from the first web.
 15. The air core reactor of claim 1, further comprising a third spacer positioned between the first cylindrical winding package and the second cylindrical winding package, the third spacer spaced circumferentially apart from the first spacer by at least a first outer flange width and comprising a third spacer end.
 16. The air core reactor of claim 15, wherein the third spacer end is disposed against an inner surface of the first outer flange.
 17. The air core reactor of claim 1, further comprising: a second attachment bracket comprising a second inner flange, a second outer flange, and a second web connecting the second inner flange to the second outer flange, the second inner flange having a second innermost surface, the second innermost surface coupled to the spider end on a spider end side opposite the first attachment bracket; a second spacer positioned between the first cylindrical winding package and the second cylindrical winding package, the second spacer having a second spacer end coupled to a second outermost surface of a second outer flange of the second attachment bracket; and a second collar disposed around the second spacer end of the second spacer between the second attachment bracket and the first cylindrical winding package and the second cylindrical winding package.
 18. The air core reactor of claim 10, wherein a second web of the second attachment bracket is attached to the first web along at least a portion of the first web height.
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